

2	(a)	(i)	sum of potential energy and kinetic energy of atoms / molecules / particles reference to random	M1	A1	[2]
		(ii)	no intermolecular forces no potential energy	B1		
			internal energy is kinetic energy (of random motion) of molecules <i>(reference to random motion here then allow back credit to (i) if M1 scored)</i>	B1		
		(b)	kinetic energy $\propto$ thermodynamic temperature <i>either</i> temperature in Celsius, not kelvin so incorrect <i>or</i> temperature in kelvin is not doubled	B1		
				B1		[2]

2 (a) temperature of the surface is the same

B1